Classical Machine Learning for Automated Sickle Cell Detection from Peripheral Blood Smear Images: A Ugandan Clinical Dataset Study

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***Abstract*—Sickle cell disease affects an estimated 13.3% of the Ugandan population, yet diagnosis remains inaccessible in rural settings due to dependence on trained laboratory personnel and specialist equipment. This study develops an automated sickle cell detection system from peripheral blood smear images captured using mobile phone cameras on basic microscopes, with the goal of enabling non-specialist health workers in rural Uganda to screen patients without laboratory expertise. Histogram of Oriented Gradients (HOG) feature extraction converts each image into an 8100-dimensional numerical feature vector. Two classifiers are trained and compared: Support Vector Machine (SVM) and Random Forest, both optimised using GridSearchCV with 5-fold cross-validation. SVM achieves the best classical ML performance with 91.98% accuracy and AUC-ROC of 0.9651, while Random Forest achieves higher sensitivity of 96.47%. Comparison with deep learning models from a prior study [2] shows that ResNet-50 achieves superior performance at 98.57% accuracy, though classical ML remains a viable alternative in low-resource settings where GPU infrastructure is unavailable. Results demonstrate that classical ML achieves clinically useful accuracy and offers a computationally accessible path to automated sickle cell screening in rural Ugandan health facilities where deep learning infrastructure is unavailable.**

***Index Terms*—Sickle Cell Disease, Machine Learning, HOG Features, SVM, Random Forest, Blood Smear Classification, Uganda, Low-Resource Healthcare**

I. Introduction

Sickle cell disease is one of the most prevalent genetic disorders in Uganda, with the sickle cell trait affecting 13.3% of the national population and reaching 19.8% in Kampala alone [1]. An estimated 15,000 to 20,000 children are born with the disease in Uganda each year. Many never receive a formal diagnosis because current diagnostic methods depend entirely on trained laboratory scientists, haemoglobin electrophoresis equipment, and manual microscopic examination of blood smears — resources concentrated in urban hospitals and largely absent from rural district health facilities. Without early diagnosis, children miss prophylactic treatment that prevents life-threatening crises, organ damage, and premature death. The diagnostic gap is not a medical problem — it is an access problem.

Automated image analysis using machine learning offers a path to closing this gap. A system that can classify a blood smear photograph taken with a mobile phone on a basic microscope could enable community health workers with no laboratory training to identify patients who need urgent referral. For such a system to be deployable in rural Ugandan health facilities, it must run on standard hardware without GPU infrastructure, which rules out deep learning in most rural settings.

Prior machine learning studies on sickle cell detection used datasets collected under controlled laboratory conditions in high-income countries. None had applied classical machine learning to the Tushabe et al. (2024) dataset — the only published dataset of sickle cell blood smear images collected from Ugandan patients using mobile phone microscopy under real clinical conditions [1]. This project addresses that gap directly.

This study applies classical machine learning with HOG feature extraction to the Tushabe (2024) dataset and compares results against deep learning models from a prior study [2]. The comparison addresses the practical question of whether classical ML can produce clinically useful results without the computational cost of deep learning.

II. Related Work

Machine learning has been applied to blood cell analysis with strong results. Acevedo et al. [3] published a dataset of 17,092 annotated peripheral blood cell images and demonstrated automated classification across eight cell types. Deep learning approaches have achieved high accuracy in medical image classification including chest X-ray analysis and blood smear interpretation [4].

For sickle cell detection, prior studies have used datasets collected under controlled laboratory conditions in high-income countries. None used mobile-phone microscopy data from clinical settings in sub-Saharan Africa until the prior study by the same author [2], which applied deep learning to the Tushabe (2024) dataset and achieved 98.57% accuracy with ResNet-50.

Poostchi et al. [5] reviewed machine learning approaches for blood smear analysis and established the feasibility of mobile-phone microscopy as an imaging platform for automated analysis, though they did not address sickle cell detection in Ugandan clinical contexts.

HOG features have been widely used for shape recognition since Dalal and Triggs [6] demonstrated their effectiveness for pedestrian detection. Their ability to capture gradient orientations makes them suited for distinguishing the morphological differences between sickle and normal red blood cells.

III. Dataset

A. Data Sources

Two datasets were combined. The primary dataset is the Tushabe et al. (2024) Ugandan Sickle Cell Microscopy Dataset [1], containing blood smear images collected from patients in Soroti and Kumi districts, Uganda, using mobile phone cameras placed on basic optical microscopes. The dataset includes 422 unlabelled sickle cell positive images and 147 normal images. An additional 422 labelled images were excluded because bounding box annotations were found burned directly into the image pixels, which would cause a classifier to learn annotation artefacts rather than cell morphology.

The supplementary dataset is the BCCD Dataset [7], providing 364 normal peripheral blood smear images captured under controlled laboratory conditions to supplement the negative class.

B. Dataset Composition

The final combined dataset contains 933 images: 422 positive (45.2%) and 511 negative (54.8%), giving a mild imbalance ratio of 1.21:1. A stratified 80/20 split produced 746 training samples and 187 test samples with class proportions preserved in both sets. Figure 1 shows the class distribution across the dataset.

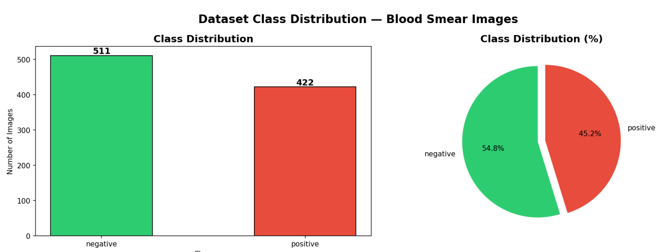

Fig. 1. Class distribution of the combined dataset showing 422 positive (sickle cell) and 511 negative (normal) images with a mild imbalance ratio of 1.21:1.

C. Data Challenges

Two data challenges were encountered and resolved. The Mendeley Peripheral Blood Cell dataset [3], originally planned as the normal class source, was found to contain segmented white blood cells rather than red blood cell smear images and was replaced with the BCCD dataset. Additionally, a domain gap exists within the negative class: Tushabe images carry a circular microscope border while BCCD images are rectangular, representing a known limitation of the combined dataset.

IV. Methodology

A. HOG Feature Extraction

HOG feature extraction converts each image into a fixed-length numerical vector by computing gradient orientation histograms in localised image regions. All images were resized to 128×128 pixels before extraction to eliminate image dimension as a potential spurious feature. HOG parameters used were 9 orientations, 8×8 pixels per cell, 2×2 cells per block, and L2-Hys block normalisation. Each image produced a feature vector of 8100 values, giving a features matrix of shape 933×8100. Figure 2 shows the HOG representation for one positive and one negative image.

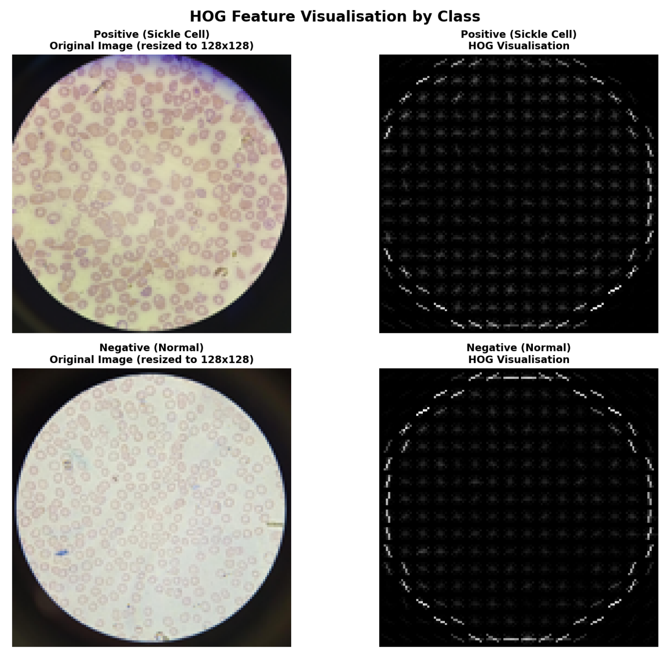

Fig. 2. HOG feature visualisation for a positive (sickle cell) and negative (normal) blood smear image after resizing to 128×128 pixels. Bright regions indicate areas of strong gradient activity corresponding to cell edges and boundaries.

B. Preprocessing Pipeline

The train-test split was performed before any preprocessing to prevent data leakage [8]. A numeric preprocessing pipeline applied StandardScaler to standardise all features to zero mean and unit variance. The scaler was fitted on training data only and used to transform both training and test sets.

C. Classifiers

Support Vector Machine (SVM): SVM finds the optimal hyperplane separating the two classes with maximum margin. The RBF kernel was used with class_weight=balanced to address class imbalance. GridSearchCV optimised C (0.1, 1, 10, 100), kernel (rbf, linear), and gamma (scale, auto) using 5-fold cross-validation with F1 scoring.

Random Forest (RF): Random Forest builds multiple decision trees on random feature subsets and combines predictions by majority vote. GridSearchCV optimised n_estimators (100, 200), max_depth (10, 20, None), min_samples_split (2, 5, 10), and min_samples_leaf (1, 2).

D. Evaluation Metrics

Models were evaluated using accuracy, AUC-ROC, F1 score, sensitivity, specificity, and precision. Sensitivity is the most clinically critical metric — missing a sickle cell patient carries greater health consequences than a false positive referral. Overfitting was assessed by comparing training and test accuracy and using 5-fold cross-validation.

V. Results

A. Classical ML Results

The optimised SVM achieved 91.98% test accuracy, AUC-ROC of 0.9651, sensitivity of 92.94%, and specificity of 91.18%, with 15 errors on 187 test images. The optimised Random Forest achieved 87.17% test accuracy, AUC-ROC of 0.9396, and sensitivity of 96.47%, with 24 errors on 187 test images. Figure 3 shows the confusion matrix and ROC curve for the optimised SVM.

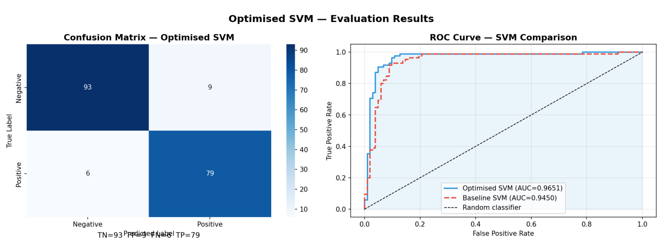

Fig. 3. Confusion matrix and ROC curve for the optimised SVM classifier. The model makes 15 errors on 187 test images with an AUC-ROC of 0.9651.

Both models showed overfitting. SVM had a training-test accuracy gap of 8.02% and cross-validation mean of 85.12%. Random Forest had a gap of 12.83% and cross-validation mean of 85.12%. GridSearchCV partially reduced the overfitting but could not fully resolve it on this small dataset.

B. Comparison with Deep Learning

Table I summarises all five models across both studies and Figure 4 shows the combined ROC curves. Deep learning results are from [2].

TABLE I
Complete Model Comparison — Classical ML vs Deep Learning

Model	Accuracy	AUC-ROC	Sensitivity
SVM (Classical ML)	91.98%	0.9651	92.94%
Random Forest (Classical ML)	87.17%	0.9396	**96.47%**
Baseline CNN (DL)*	87.86%	0.9588	95.24%
EfficientNet-B0 (DL)*	96.43%	0.9951	95.24%
ResNet-50 (DL)*	**98.57%**	**0.9973**	98.41%

*Results from [2]

Deep learning with transfer learning clearly outperforms classical ML on this task. ResNet-50 achieves 98.57% accuracy compared to SVM at 91.98%, a gap of 6.59 percentage

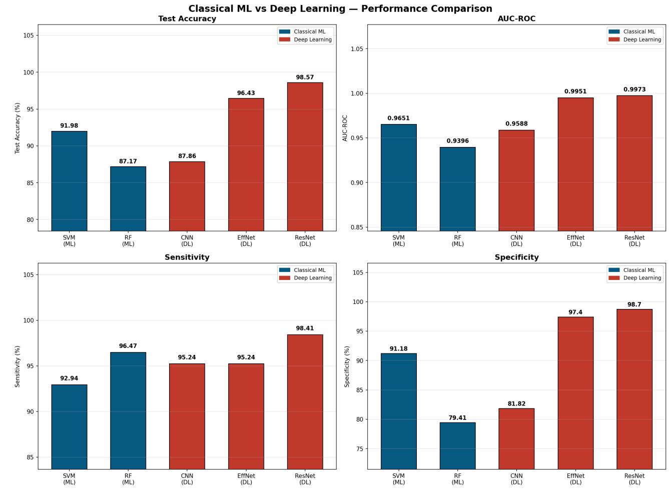

Fig. 4. Performance comparison across all five models showing test accuracy, AUC-ROC, sensitivity, and specificity. Deep learning models are shown in red and classical ML models in blue.

points that is clinically meaningful. The one area where classical ML is competitive is Random Forest sensitivity at 96.47%, which exceeds EfficientNet-B0 at 95.24%.

VI. Discussion

Deep learning significantly outperforms classical ML on this task. ResNet-50 achieves 98.57% accuracy and AUC-ROC of 0.9973, compared to SVM at 91.98% and AUC-ROC of 0.9651. The 6.59% accuracy gap is clinically meaningful — deep learning should be the preferred approach wherever GPU infrastructure is available.

Deep learning outperforms classical ML primarily because it learns features directly from raw images through multiple convolutional layers, automatically discovering discriminative patterns without human design. Transfer learning from ImageNet gives EfficientNet-B0 and ResNet-50 rich visual representations that HOG features cannot replicate.

Classical ML has meaningful practical advantages in the Ugandan deployment context. It requires no GPU infrastructure, trains in minutes, and achieved over 90% accuracy with SVM. In rural Ugandan health facilities where GPU hardware is unavailable, SVM with HOG features represents a computationally accessible alternative that still achieves clinically useful accuracy and could run on a basic laptop or mobile device, directly supporting point-of-care screening by non-specialist health workers.

VII. Limitations

The dataset of 933 images is small, limiting model generalisation. Both classical ML models show overfitting that was not fully resolved through hyperparameter tuning. The domain gap between Tushabe mobile-phone images and BCCD laboratory images introduces visual inconsistency within the negative class. Comparison with deep learning results uses a different test set size (187 vs 140 samples). No external clinical validation was performed.

VIII. Conclusion

This study developed an automated sickle cell detection system using classical machine learning applied to the Tushabe (2024) Ugandan clinical mobile-phone microscopy dataset — the first study to apply classical ML to this dataset. SVM achieved 91.98% accuracy and AUC-ROC of 0.9651, demonstrating that classical ML can produce clinically useful results for sickle cell screening without GPU infrastructure.

Comparison with deep learning models from a prior study shows that ResNet-50 significantly outperforms classical ML at 98.57% accuracy, and deep learning should be the preferred approach wherever computational resources allow. However in low-resource Ugandan health facilities where GPU infrastructure is absent, SVM with HOG features offers a viable and deployable alternative that could meaningfully improve access to sickle cell screening for patients who currently receive no diagnosis at all.

Future work should address overfitting through larger datasets, explore additional feature extraction methods, and conduct clinical validation in Ugandan health facilities.

References

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